AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

M348
Applied Statistical Modelling

Final Examination
Spring Semester 2024/2025

Date: 11/June/2025

Number of Exam Pages: 7 (including this cover sheet)	**Time Allowed: 3 hours**

Instructions:

- Total Marks: **50**
- This exam consists of **4** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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PART 1: True or False Questions [20 x 0.5=10 marks]

Which of the following statements are true/false?

1. AIC (Akaike Information Criterion) is used to compare models, and lower values indicate a better fit.
2. In logistic regression, the link function is typically linear.
3. Logistic regression is not appropriate when the response variable is continuous.
4. In multiple regression, the R-squared value indicates the proportion of variance in the response variable explained by the predictors.
5. In a simple linear regression model, the regression line is fitted using the least squares method.
6. In linear regression, the residuals should be dependent of the predicted values.
7. In multiple regression, a low p-value for a predictor variable indicates that it is a statistically significant predictor.
8. In regression, multicollinearity occurs when independent variables are highly correlated with each other.
9. In a multiple regression model, the null hypothesis (H0) is that all regression coefficients are not zero.
10. The odds ratio in logistic regression indicates how much the odds of success increase for each unit change in the predictor variable.
11. The t-statistic in regression testing helps us assess if the coefficient of a predictor variable is significantly different from zero.
12. In regression models, interaction terms are used to model the combined effect of two predictors on the response variable.
13. In a model with non-parallel slopes, the interaction term indicates that the effect of one predictor on the response variable depends on the level of another predictor.
14. The standard error of the coefficient is used to calculate the confidence interval for the coefficient.
15. The residual standard error provides an estimate of the standard deviation of the residuals.
16. If the residuals are not normally distributed, the model's predictions may be unreliable.
17. The F-statistic in regression analysis tests if at least one predictor variable has a significant relationship with the response variable.
18. In a multiple regression model, including interaction terms always improves the model fit.
19. In regression, the inclusion of factor variables requires coding them into numerical variables (e.g., dummy variables).
20. In ANOVA, we compare the residual sum of squares (RSS) for different models to determine which model fits the data better.

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PART 2: Multiple linear regression for the OU students dataset [13 x 1= 13 marks]

ALL QUESTIONS ARE REQUIRED

The Open University Students Dataset provides insights into students' academic performance and socio-economic background. It includes various features related to students' demographics, prior academic achievements, and assessment scores. This dataset can be valuable for analyzing factors influencing students' success and identifying patterns in their performance.
The dataset consists of 9 variables, categorized as numerical and categorical features:

- modResult: A binary variable (1 or 0), possibly indicating whether the student successfully completed the module.
- examScore: The final score obtained in the module's exam.
- contAssScore: The score obtained in continuous assessments throughout the course.
- bestPrevModScore: The best previous module score achieved by the student.
- age: The student's age at the time of enrollment.
- region: The geographical region where the student is located (e.g., "West Midlands", "London").
- gender: The gender of the student (m for male, f for female).
- imd: The Index of Multiple Deprivation (IMD), representing the student's socio-economic background (e.g., "most deprived", "least deprived", "middle deprived", "other").
- qualLink: Indicates whether the student's previous qualifications were related to the current module (e.g., "maths", "not").

This dataset is useful for predicting academic performance, understanding socio-economic impacts on education, and analyzing learning outcomes at the Open University.

1) Complete the code below to fit a multiple regression model for the response variable examScore using only the numerical explanatory variables: contAssScore, bestPrevModScore, and age. Store the results in an object called examScoreFit. Obtain a summary of the results from fitting the model.

```
a. examScoreFit <- lm()
b. summary(   )
```

The resulting output are given below:

```
Residuals:
    Min      1Q  Median      3Q     Max
-59.095  -8.337   1.828   9.845  46.283
```


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```
Coefficients:
                Estimate Std. Error t value Pr(>|t|)
(Intercept)     -17.23451   3.51647  -4.901 1.04e-06 ***
contAssScore      0.65724   0.02328  28.232  < 2e-16 ***
bestPrevModScore  0.47139   0.04440  10.618  < 2e-16 ***
age              -0.14915   0.02810  -5.307 1.25e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 13.95 on 1792 degrees of freedom
Multiple R-squared:  0.4631,    Adjusted R-squared:  0.4622
F-statistic: 515.3 on 3 and 1792 DF,  p-value: < 2.2e-16
```

2) What is the fitted regression equation for the model?
3) Interpret the coefficient of each explanatory variable (contAssScore, bestPrevModScore, and age). What does the negative coefficient of age (-0.14915) imply about its relationship with examScore?
4) What does the R-squared value (0.4631) indicate about the model's performance?
5) What does the p-value for each coefficient tell us about its significance?
6) Using the standard error of bestPrevModScore (0.04440), construct a 95% confidence interval for its coefficient.
7) What is the estimated standard deviation (σ) of the residuals, and what does it tell us about model accuracy?
8) Given a student with:
   - contAssScore = 75
   - bestPrevModScore = 85
   - age = 30

   What is the predicted examScore?
9) How can you visually assess whether residuals follow a normal distribution?
10) The residual standard error in this model is 13.95, while a simple regression model had an error of 20. How does this affect our model's performance?
11) What steps would you take to determine if this model is overfitting?
12) How would you use stepwise regression to find a better model?
13) What trade-offs exist between a model's complexity (flexibility) and interpretability?

PART 3: Parallel slopes model for the OU students dataset [19x 1= 19 marks]

ALL QUESTIONS ARE REQUIRED

1) Adapt the regression model to fit a parallel slopes model, where the response variable is examScore, and the explanatory variables are gender, imd, bestPrevModScore, and age.
   - Store the results in an object called ParallelFit.

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- Use the summary() function to display the model output.

The resulting output of part (1) are given below:

```
Residuals:
    Min      1Q  Median      3Q     Max
-72.963  -9.966   2.808  12.308  55.180

Coefficients:
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)     -26.61613    4.27896  -6.220 6.17e-10 ***
genderm           0.58997    0.82578   0.714  0.47505
imdmiddle         1.30001    1.12289   1.158  0.24713
imdleast          1.17997    1.05871   1.115  0.26520
imdother          3.42250    1.52191   2.249  0.02465 *
bestPrevModScore  1.06818    0.04723  22.618  < 2e-16 ***
age              -0.09190    0.03391  -2.710  0.00678 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 16.76 on 1789 degrees of freedom
Multiple R-squared:  0.2269,    Adjusted R-squared:  0.2243
F-statistic:  87.5 on 6 and 1789 DF,  p-value: < 2.2e-16
```

2) Based on the output, which gender has R taken as the reference (level 1) for the gender variable?
3) Using the summary output, interpret the estimated value of the intercept in the context of the model (gender: M or F, imd: most, least, middle, other).
4) What does the estimated coefficient of genderm represent in this model?
5) Is there evidence to suggest that gender has a statistically significant effect on examScore? Justify your answer using the p-value.
6) What does the coefficient for bestPrevModScore suggest about its impact on examScore?
7) What does the coefficient for imdother suggest about students from that group compared to the baseline IMD category?
8) If a model were fitted without imd, would you expect the adjusted R^2 to increase or decrease? Explain why.
9) Based on the F-statistic and its associated p-value, is the overall model statistically significant?
10) Why is the Adjusted R-squared slightly lower than the Multiple R-squared, and what does it indicate?
11) What is the predicted examScore for a male student with an imd category of middle, bestPrevModScore = 60, and age = 25?
12) What is the predicted examScore for a female student with an imd category of least, bestPrevModScore = 75, and age = 22?
13) If we increase a student's age by 5 years, how would their predicted examScore change, assuming other factors remain constant?

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14) What additional insights might we gain if we included an interaction term (gender * bestPrevModScore)?

15) If the p-value for gender were above 0.10, would you recommend removing gender from the model? Why or why not?

To determine whether gender should be included in the model, we first fit a reduced model without gender and compare it to the full model using the anova() function. We then perform a similar test for imd to assess its importance in the model

After removing gender from the model, the ANOVA test results are shown below. This test evaluates whether gender contributes significantly to explaining the variability in the response variable:

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1792	647071.2	NA	NA	NA	NA
1791	646988.6	1	82.64412	0.2287763	0.6324911

After removing imd from the model, the ANOVA test results are shown below. This test evaluates whether imd has a significant effect on the response variable:

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
1794	649578.2	NA	NA NA	NA	
1791	646988.6	3	2589.673	2.389586	0.06707499

16) What does the p-value tell us about whether imd should be included in the model (second Anova table)?

17) What does the p-value tell us about whether gender should be included in the model?

18) What is the null hypothesis being tested in each of the ANOVA comparisons.

19) How can we apply the anova() function to test whether gender should be included in the model? Provide the corresponding R code to perform this test.

PART 4: Generalized linear model for the OU student's dataset [8 x 1= 8 marks]

ALL QUESTIONS ARE REQUIRED

We utilized the glm() function in R to fit a logistic regression model for the OU dataset, with modResult as the response variable and age, gender, imd, and bestPrevModScore as explanatory variables. The modResult variable is described as follows:

- modResult: A binary variable where 1 indicates the student passed the exam and 0 indicates they did not.

1) Complete the code below to fit the logistic regression model modResult in function of age, gender, imd and bestPrevModScore, storing the result in an object called LogisticFit, and produce a summary of the fitted model.

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- LogisticFit<- glm(,)
- summary()

The resulting output of part (1) are given below:

```
Coefficients:
                  Estimate Std. Error z value Pr(>|z|)
(Intercept)     -4.558374   0.805247  -5.661 1.51e-08 ***
age             -0.014576   0.009606  -1.517    0.129
genderm         -0.062889   0.249445  -0.252    0.801
imdmiddle        0.014164   0.321374   0.044    0.965
imdleast         0.007830   0.310147   0.025    0.980
imdother         0.502823   0.472398   1.064    0.287
bestPrevModScore 0.093347   0.009440   9.888  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 678.51  on 1795  degrees of freedom
Residual deviance: 585.02  on 1789  degrees of freedom
AIC: 599.02

Number of Fisher Scoring iterations: 6
```

2) What is the equation of the fitted logistic regression model?
3) What are the key differences between logistic regression and linear regression, and why is logistic regression used in this case?
4) What does the p-value for age (0.129) suggest about its statistical significance in predicting the exam outcome?
5) Based on the residual deviance and null deviance, how well does the model explain the variation in the modResult variable?
6) If a student is 25 years old, male, has a best previous module score of 85, and is from the 'least' IMD category, what is the predicted probability of passing the exam?
7) What does the coefficient for bestPrevModScore represent in terms of log-odds in this logistic regression model?
8) In the context of generalized linear models (GLMs), what are the canonical link functions for the following distributions?
   - Poisson
   - Bernoulli
   - Binomial
   - Exponential

End of questions

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